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Project Title:	Pharmacological Intervention to Prevent Disuse Osteopenia		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Zip Code:	97331-8558	**Congressional District:**	4
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	We have previously obtained evidence suggesting that mechanical loading acts on bone cells via a cascade mechanism. We have postulated that the initial step of the cascade initiated by dynamic loading involves mechanoreceptor-transduced regulation of expression of a small number of immediate response genes which code for transcription factors and other cell cycle regulatory factors. These factors regulate expression of genes which code for skeletal signaling molecules (e.g., growth factors) which in turn influence the expression of genes to control cell proliferation, metabolism and expression of differentiated bone cell function. If this model is correct, disuse osteopenia is largely mediated by altered expression of cell signaling molecules and may be corrected by pharmacological agents which modulate the skeletal expression of these intermediates. Studies performed during the previous funding interval have revealed that the signal transduction pathways for parathyroid hormone (PTH) and estrogen partially overlap with		

Task Description:	weight bearing in that each of these agents regulates the expression of skeletal signaling molecules. These studies also demonstrated an unexpected pronounced gender difference in the skeletal response to decreased mechanical usage. The proposed research is designed to test the hypothesis that it is possible to bypass the mechanoreceptor step in hindlimb unloaded rats by pharmacological intervention with PTH and a tissue-selective estrogen agonist (clomiphene) with little activity on reproductive tissues. The specific aims of the proposed studies in hindlimb elevated rats are to determine: 1) the comparative effects of skeletal unloading on normal and gonadectomized males and females; 2) the efficacy of clomiphene treatment to prevent cancellous osteopenia and maintain normal bone turnover, architecture and mechanical properties; and 3) if the skeletal efficacy of combined treatment with clomiphene and human PTH (hPTH) is superior to monotherapy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although these studies were directed at developing a method to prevent bone loss during spaceflight, the results are highly relevant to human disease. In the course of these studies, we identified PDGF-A as a putative causative factor for parathyroid hormone bone disease and established that antagonism of PDGF-A signaling can greatly reduce hypercalcemia, focal bone resorption, and osteitis fibrosa in an animal model for hyperparathyroidism. If confirmed in humans, drugs that target PDGF-A signaling could be used to treated patients with renal osteodystrophy as well as individuals with primary and other forms of secondary hyperparathyroidism. We have also shown that "bone specific" genes are expressed in white blood cells and respond in an analogous manner to PTH as in bone and kidney. These findings initially suggested that nucleated blood cells can be used as a surrogate tissue to monitor positive as well as adverse responses to systemic factors in organs which cannot be directly evaluated without use of invasive procedures. Recent studies suggest that these cell represent preosteoblasts and are likely to play a significant role in normal bone metabolism as well as metabolic bone disease and disuse.
Task Progress:	All of our studies were related to Risk 1: “Accelerated bone loss and fracture risk”, in the Critical Path Roadmap. Question 1c: “Is there an additive or synergistic effect of gonadal hormone deficiency in men or women on bone loss during prolonged exposure to hypogravity?” Our studies have helped to answer this question by showing that: Estrogen deficiency and hypogravity have additive effects on bone loss in rats. Also, skeletal unloading resulted in bone loss in males and females by gender-independent mechanisms, increased bone resorption and decreased bone formation in both. However, our results also demonstrated that most of the bone loss in rats subjected to microgravity is due to decreased bone formation in animals with normal estrogen levels whereas it is due to increased bone resorption in estrogen deficient rats. As a consequence, the ideal drug (or exercise) intervention may have to take into consideration the gonadal hormone status individual astronauts. There is compelling evidence that estrogen sensitizes the skeleton to mechanical loading. We have investigated the possible contributions of metabolites of 17ß-estradiol, to the estrogen effect, especially those which are important following the menopause. These studies have revealed that 16?-hydroxyestrone is a complete estrogen agonist on bone but a partial agonist on other estrogen target tissues. We demonstrated gender differences in bone mass and bone turnover in adult rats. These differences (e.g., females have a higher rate of bone turnover than males) are similar to what we have observed in human iliac crest bone biopsies. Future studies will be needed to establish whether there are gender specific differences in the rates of bone loss. Additionally, studies should be performed to establish whether there are gender differences in the rates of recovery during hypogravity. Question 1d: “What pharmacological agent(s) will most effectively minimize the decrease in bone mass with extended exposure to hypogravity?” We have helped answer this question by performing experiments which have shown that intermittent PTH can completely prevent the pronounced inhibitory effects of hindlimb microgravity on bone formation as well as prevent bone loss. We are continuing to investigate whether combining PTH with SERMs has additional beneficial effects on the skeleton during hypogravity. We have investigated further the mechanisms mediating the detrimental side effects of PTH. This is important because side effects are likely to occur in some individuals receiving PTH therapy to prevent bone loss and the incidence of side effects might occur during hypogravity. We obtained evidence that the catabolic bone effects of PTH require PDGF-A signaling. Antagonizing this signaling with an inhibitor of PDGF-A expression and receptor binding blunted PTH-induced bone marrow fibrosis, hypercalcemia and bone resorption. . Question 1h: “Does the hypogravity environment change the nutritional requirements for optimal bone health?” We have obtained data that strongly suggest that the hypogravity environment changes the response to “life style factors” and nutritional factors. We have investigated how selected “life style factors” such as alcohol consumption alter the skeletal response to mechanical unloading. These studies demonstrated that alcohol and hindlimb unloading have independent detrimental effects on bone formation. It is unlikely that astronauts will be consuming alcohol during spaceflight. However, our data suggests that detrimental skeletal effects of consuming alcohol persist when drinking ceases and could potentially exaggerate the effects of hypogravity. We have compelling evidence that weight bearing and PTH act in synergy to increase bone formation. The level of PTH in blood is greatly influenced by ionic calcium. As a consequence, the circulating levels of PTH during hypogravity may be inadequate to maintain normal bone formation. Question 1j: “What systemic adaptation s to hypogravity are important contributing factors to bone loss, evaluations of which are essential to effective countermeasure development?” This question was not an original goal of the proposed research. However, we have now accumulated strong evidence that the mast cell functions as a critical accessory cell in mediating the detrimental skeletal effects of hyperparathyroidism, estrogen deficiency and inflammation. Our future efforts will include investigating the potential

	role of mast cells in mediating hypogravity-induced bone loss. As a starting point, we will investigate the relationship between mast cell number in bone in response to hypogravity using archival studies. Depending on the results, future studies will be performed in mast-cell deficient animals. These studies have the potential to reveal a novel target for the development of pharmacological agents to prevent disuse osteopenia. Question 1k: “Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity?” We have performed additional studies to further develop our model systems and ability to accurately measure bone density, geometry and architecture. These include establishing the precise age of epiphyseal closure in rats using micro-CT and developing cryostatic micro-CT imaging. Using the latter method, we are able to obtain high resolution images for measuring in 3-D geometry and architecture while preserving the integrity of RNA within the tissue. We have established a method whereby we can obtain RNA following imaging in order to investigate gene expression in precise regions of bone. Our future efforts will focus on establishing the skeletal changes resulting from a return to full gravity following prolonged microgravity. Question 1l: “What regimen will most effectively hasten restoration of bone mass and bone strength to pre-flight values in returning crewmembers?” Year 3 of the grant will focus on this question. [Ed. note: continues as grant number NNJ05HD45G with PI move to Oregon State University.]
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Lotinun S, Evans GL, Turner RT, Oursler MJ. "Deletion of membrane-bound steel factor results in osteopenia in mice." J Bone Miner Res. 2005 Apr;20(4):644-52. PMID: 15765184 , Apr-2005